

Elementary Functions

1. Dirichlet Function

$D: \mathbb{R} \rightarrow \{0, 1\}$ is given by $D(x) = 1$ if $x \in \mathbb{Q}$ and $D(x) = 0$ if $x \in \mathbb{I}$

2. Absolute value Function

$f: \mathbb{R} \rightarrow [0, \infty)$ is given by $f(x) = x$ if $x \geq 0$ and $f(x) = -x$ if $x < 0$
 $f(x) = |x|$

3. Sign Function

$f: \mathbb{R} \rightarrow \{-1, 0, 1\}$ is given by $f(x) = \text{sign}(x)$

if $x > 0$, $f(x) = 1$

if $x = 0$, $f(x) = 0$

if $x < 0$, $f(x) = -1$

4. Heaviside Function

$f: \mathbb{R} \rightarrow \{0, 1\}$ is given by

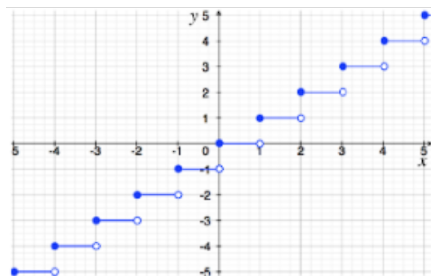
$H(x) = 1$ if $x > 0$

$H(x) = 0$ if $x \leq 0$

5. Integer part Function (Floor Function)

The integer part of a number $x \in \mathbb{R}$, $|x|$ is the largest integer $z \in \mathbb{Z}$ such that $z \leq x$

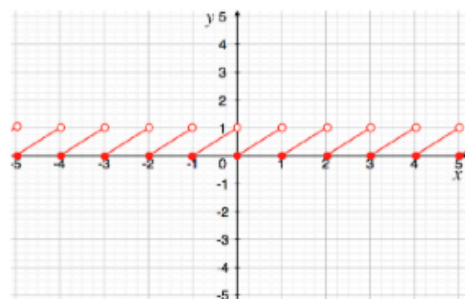
(For example, $[2.5] = 2$ or $[-2.5] = -3$)



6. Fractional part Function

The fractional or decimal part of a number $x \in \mathbb{R}$ is defined as $\{x\} = x - \lfloor x \rfloor$

(For example, $\{2.5\} = 0.5$ or $\{-2.3\} = 0.7$)



7. Linear Functions

Linear Functions have the form $y = f(x) = mx + b$

$\frac{y_2 - y_1}{x_2 - x_1}$ is the gradient (slope)

b is the y-intercept

$P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are any 2 points on the graph of f

8. Power Functions

Power Functions are given by the rule $f(x) = x^n$, where $n \in \mathbb{R}$

- If $n \in \mathbb{N}$
 - $f(x)$ is even when n is even $\rightarrow f(-x) = f(x)$
 - $f(x)$ is odd when n is odd $\rightarrow f(-x) = -f(x)$
- If $n \in \mathbb{Q}$ and $n = \frac{p}{q}$, p, q mutually prime
 - $f(x)$ is defined over all the real numbers if q is odd, but only on $[0, \infty)$ if q is even
 - If q is odd, $f(x)$ is even when p is even, and odd when p is odd
- If n is irrational and positive. In this case, $f(x)$ is defined only for $x > 0$
 - Examples: $f(x) = x^{\sqrt{2}}$ or $f(x) = x^\pi$
 - $x^{\sqrt{2}} = e^{\sqrt{2}\ln(x)}$ accepts only positive values of x
- If $n \in \mathbb{R}$ and $n < 0$, all properties before apply but the function is not defined in zero

9. Polynomial Functions

A polynomial function $P: \mathbb{R} \rightarrow \mathbb{R}$ of degree $n \in \mathbb{N}$ is defined as

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

- $a_i \in \mathbb{R}$ are constants, for $i = 0, 1, 2, \dots, n$. These constants can be 0 except for the term corresponding to $i = n$
- The degree of the polynomial corresponds to the highest term x^n in the expression

10. Rational Functions

$$R(x) = \frac{P(x)}{Q(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0}{a_m x^m + a_{m-1} x^{m-1} + \dots + a_2 x^2 + a_1 x + a_0}$$

- Generally $n \neq m$
- Proper Rational Function: Degree of $P(x)$ is less than the degree of $Q(x)$
- Improper Rational Function: Degree of $P(x)$ is greater than the degree of $Q(x)$
- Improper rational functions can always be rewritten as the sum of a polynomial $S(x)$ and a proper rational function $\frac{T(x)}{Q(x)}$

11. Algebraic Functions

A function $f: X \rightarrow Y$ is said to be algebraic, if it satisfies the algebraic equation:

$$P_n(x)y^n + P_{n-1}(x)y^{n-1} + \dots + P_2(x)y^2 + P_1(x)y + P_0 = 0$$

for $n \in \mathbb{N}$ where $P_0(x), \dots, P_n(x)$ are polynomials

12. Exponential and Logarithm Functions

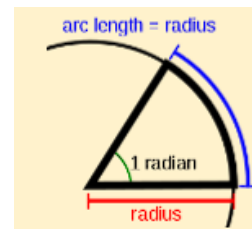
An exponential function is a function of the form $f(x) = a^x$ where $a > 0$ is a constant ($a \neq 1$). Its inverse is the logarithm function with base a , indicated as $g(x) = \log_a x$

- The exponential is always positive. It is increasing for $a > 1$, decreasing for $0 < a < 1$
- If the exponential is a number y given by $y = a^p$, the logarithm to base a of y is p ($\log_a y = p$)
- The logarithm is defined only for positive values of its argument
- If $a > 1$:
 - The logarithm is always increasing
 - $\log_a x \geq 0$
 - $\log_a x < 0$ if $0 < x < 1$
- If $0 < a < 1$, the opposite statements are true
- The most common bases are 10 and e

13. Trigonometric Functions (Periodic / Circular Functions)

A function $f: X \rightarrow Y$ is periodic with period T , if there exists a number $T \in \mathbb{R}$ such that, for every $x \in X$ we have $f(x + T) = f(x)$

- The $\sin \theta$ is the ordinate of P
- The $\cos \theta$ is the abscissa of P
- The $\tan \theta = \frac{\sin \theta}{\cos \theta}$



Important points:

- A positive angle θ is measured in a counter-clockwise fashion
- The angle is measured in radians. 1 rad is the angle corresponding to the arc whose length is equal to the radius of the circle. Hence, the length of an arc of a circle making an angle θ is simply $s = r\theta$

Properties:

- The area of a sector of a circle of angle θ is $A = \frac{1}{2}r^2\theta$
- Sine, Cosine and Tangent are periodic, with period $T = 2\pi$ for sine and cosine and $T = \pi$ for tangent
- Sine and Tangent are odd functions while Cosine is an even function
- Cotangent: $\cot\theta = \frac{\cos\theta}{\sin\theta}$
- Secant: $\sec\theta = \frac{1}{\cos\theta}$
- Cosecant: $\csc\theta = \frac{1}{\sin\theta}$
- $\sec^2\theta - \tan^2\theta = 1$
- Important formulae:
 - $\sin(x + y) = \sin x \cos y + \sin y \cos x$
 - $\sin(x - y) = \sin x \cos y - \sin y \cos x$
 - $\cos(x + y) = \cos x \cos y - \sin x \sin y$
 - $\cos(x - y) = \cos x \cos y + \sin x \sin y$

Inverse Trigonometric functions:

- $y = \sin^{-1}x = \arcsin x$, $\text{dom}(\arcsin x) = [-1, 1]$, $\text{ran}(\arcsin x) = [-\frac{\pi}{2}, \frac{\pi}{2}]$
- $y = \cos^{-1}x = \arccos x$, $\text{dom}(\arccos x) = [-1, 1]$, $\text{ran}(\arccos x) = [0, \pi]$
- $y = \tan^{-1}x = \arctan x$, $\text{dom}(\arctan x) = \mathbb{R}$, $\text{ran}(\arctan x) = (-\frac{\pi}{2}, \frac{\pi}{2})$
- The arcsine and arctangent are odd functions. The arccosine is neither odd nor even

14. Hyperbolic Functions:

Important points:

- $\cosh x = \frac{1}{2}(e^x + e^{-x})$
- $\sinh x = \frac{1}{2}(e^x - e^{-x})$
- $\tanh x = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$

Properties:

- $\cosh^2 x - \sinh^2 x = 1$
- $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$
- $\cosh(x - y) = \sinh x \cosh y - \cosh x \sinh y$
- $\sinh(x + y) = \cosh x \cosh y + \sinh x \sinh y$
- $\sinh(x - y) = \cosh x \cosh y - \sinh x \sinh y$
- $\coth x = \frac{\cosh x}{\sinh x}$
- $\text{sech } x = \frac{1}{\cosh x}$
- $\text{csch } x = \frac{1}{\sinh x}$
- $\text{sech}^2 x + \tanh^2 x = 1$

Inverse Hyperbolic Functions:

- $\cosh^{-1}x = \operatorname{arccosh}x = \ln(x + \sqrt{x^2 - 1})$
 $\operatorname{Dom}(\operatorname{arccosh}x) = [1, \infty)$, $\operatorname{Ran}(\operatorname{arccosh}x) = [0, \infty)$
- $\sinh^{-1}x = \operatorname{arcsinh}x = \ln(x + \sqrt{x^2 + 1})$
 $\operatorname{Dom}(\operatorname{arcsinh}x) = \mathbb{R}$, $\operatorname{Ran}(\operatorname{arcsinh}x) = \mathbb{R}$
- $\tanh^{-1}x = \operatorname{arctanh}x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$
 $\operatorname{Dom}(\operatorname{arctanh}x) = (-1, 1)$, $\operatorname{Ran}(\operatorname{arctanh}x) = \mathbb{R}$

The concept of Limit

For $a, L \in \mathbb{R}$, we say that a function $f(x)$ has the limit L as x approaches a :

$$\lim_{x \rightarrow a} f(x) = L$$

If we can make the value of $f(x)$ arbitrarily close to L for all x in a sufficiently small neighbourhood of a (x may be either side of a but not equal to a)

Notes:

- The limit can exist even if the function is not defined at $x = a$
- The limit can exist even if the function has a different value at $x = a$
- If we cannot make the value of the function arbitrarily close to some value L , we say that the limit does not exist

One-sided limits:

- Given $a, L_1 \in \mathbb{R}$, we say that x approaches a from the left-hand side when only values of $x < a$ are considered in the limit. If the result of the operation is L_1 then we say:

$$\lim_{x \rightarrow a^-} f(x) = L_1$$

- Given $a, L_2 \in \mathbb{R}$, we say that x approaches a from the right-hand side when only values of $x > a$ are considered in the limit. If the result of the operation is L_2 then we say:

$$\lim_{x \rightarrow a^+} f(x) = L_2$$

- The only way for a limit to exist at a given point $a \in \mathbb{R}$ is that both left and right-hand sided limits exist and are equal. Given $a, L \in \mathbb{R}$:

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if} \quad \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

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- **Limits Laws:**

For 2 generic functions $f(x)$ and $g(x)$, given $a, L_1, L_2 \in \mathbb{R}$ and supposing that

$$\lim_{x \rightarrow a} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = L_2$$

- $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = L_1 \pm L_2$
- $\lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x) = cL_1$
- $\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x) = L_1L_2$
- $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2} \quad \text{if } L_2 \neq 0$
- $\lim_{x \rightarrow a} c = c$
- $\lim_{x \rightarrow a} x = a$